

K-Free Dependence Bayesian Classifiers

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Abstract—As one of the most attractive Bayesian network classifiers (BNCs), the K-dependence Bayesian (KDB) classifier can effectively capture dependencies between attributes by allowing each of them to be conditioned on the class and, at most, K other attributes. However, as K becomes larger, its structural complexity greatly increases, which inevitably leads to a certain risk of overfitting. Moreover, when K is given, its structure is immutable, which dramatically limits the expression ability of the final model. To address these two issues, in this study, we propose K-free dependence Bayesian (KFDB) classifiers, which can learn an adaptive number of parent nodes for each attribute. To search its optimal structure, we sequentially evaluate the candidate submodels either by minimizing the mean squared error (MSE) or maximizing the classification accuracy (ACC), resulting in two versions denoted as KFDB^{MSE} and KFDB^{ACC}, respectively. Experimental results on 60 benchmark datasets demonstrate that KFDB significantly outperforms the classical KDB and other state-of-the-art models.

Index Terms—Attribute dependency, classification, K-dependence Bayesian (KDB) classifiers, structure learning.

I. INTRODUCTION

CLASSIFICATION is one of the most fundamental tasks in machine learning and data mining [1], [2], [3]. Over

the past few decades, numerous classifiers have been introduced, and they have been successfully applied to a wide range of real-world applications, such as image classification and object detection [4], [5]. Compared with other classifiers, Bayesian network classifiers (BNCs) [6], [7] receive much attention due to their obvious advantages, such as the explicit interpretability and the powerful expression ability. A BNC is a type of Bayesian network designed for classification tasks, defined by two parameters: $\mathcal{G}$ and Θ . $\mathcal{G}$ denotes the directed acyclic graph in which the nodes correspond to attributes and the arcs represent the dependencies between attributes, while Θ denotes the conditional probability tables that quantify the probabilistic model. Given a test instance $\mathbf{x}$ represented by an attribute value vector $\langle a_1, a_2, \dots, a_m \rangle$, BNC uses (1) to predict its class label

$$c(\mathbf{x}) = \arg \max_{c \in C} P(c) P(a_1, a_2, \dots, a_m | c) \quad (1)$$

where $c(\mathbf{x})$ is the predicted class label of $\mathbf{x}$, $P(c)$ is the prior probability of the class c , $P(a_1, a_2, \dots, a_m | c)$ is the joint probability given the class c , m is the number of attributes, a_m is the m th attribute value of $\mathbf{x}$, and C is the collection of all possible class labels c .

Assume that all attributes are fully independent given the class, the resulting BNC is called naive Bayes (NB). Despite its simple structure, NB has demonstrated remarkable classification performance and continues to be one of the top 10 data mining algorithms [8]. However, its attribute conditional independence assumption rarely holds true in real-world applications. To mitigate this unrealistic assumption, numerous enhancements have been proposed, which could be broadly divided into three categories: structure-oriented improvements, instance-oriented improvements, and attribute-oriented improvements. Among them, structure-oriented improvements [9], [10], [11] attempt to introduce additional directed arcs to explicitly represent attribute dependencies, relaxing the attribute conditional independence assumption in NB. Instance-oriented improvements [12], [13], [14] mainly take into account the discriminative contribution of different instances on the estimation of prior and conditional probabilities, thereby improving the accuracy of probability estimation. Attribute-oriented improvements [15], [16], [17] account for the varying importance of different attributes in model construction and relax the conditional independence assumption from the attribute perspective.

Compared to other enhancements, structure-oriented improvements exhibit several distinctive advantages in terms of interpretability, scalability, and performance. First, they provide a graphical dependency structure that offers intuitive insights into the decision-making process with explicit

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interpretability. Second, they enable the insertion or deletion of nodes and arcs within the existing network structure, allowing dynamic updates without reconstructing the entire model, which ensures strong scalability. Third, they employ directed arcs to represent attribute dependencies, thereby capturing more complex data characteristics and achieving superior classification performance. Therefore, we mainly concentrate on structure-oriented improvements in this study.

However, learning the optimal structure of BNC is NP-hard [18], as exhaustively exploring all possible dependencies between attributes is computationally infeasible. To address this challenge, restricted BNCs have been proposed, which can be further categorized into one-dependence estimators (ODEs) [19], [20], [21] and K-dependence estimators (KDEs) [22], [23], [24]. In ODE, each attribute is conditioned on the class and up to one other attribute, whereas in KDE, each attribute can depend on the class and, at most, K other attributes. Compared with ODE, KDE generally captures attribute dependencies in a more comprehensive manner, leading to enhanced model expressiveness. The most representative KDE is called K-dependence Bayesian (KDB). However, to the best of our knowledge, the value of K in KDB is typically pre-determined by the user, and it is usually difficult to obtain sufficient prior knowledge to select an appropriate K for specific applications. On one hand, as K becomes larger, the complexity of KDB grows significantly, which inevitably leads to a certain risk of overfitting. On the other hand, when K is given, the structure of KDB becomes immutable, i.e., the dependencies between attributes are fixed, which dramatically limits the expression ability of the final model.

Therefore, in this study, we propose K-free dependence Bayesian (KFDB) classifiers, which can learn an adaptive number of parent nodes for each attribute. In KFDB, we first greedily learn the structure of multiple candidate submodels based on a pre-set K, and then, we sequentially evaluate them using incremental leave-one-out cross-validation (LOOCV) by minimizing the mean squared error (MSE) or maximizing the classification ACC. We denote the resulting model as KFDB^{MSE} and KFDB^{ACC}. To validate their effectiveness, we conduct extensive experiments on a collection of 60 benchmark datasets from the University of California at Irvine (UCI) repository. Experimental results show that KFDB significantly outperforms the classical KDB and other state-of-the-art models.

In summary, the main contributions of this article are presented as follows.

- 1) We identify a new problem that although the value of K is a crucial hyperparameter in KDB, determining an appropriate K value is very challenging in real-world applications. Too large K will increase the risk of overfitting, while a fixed K will make the structure immutable.
- 2) We propose KFDB classifiers, which can learn an adaptive number of parent nodes for each attribute by sequentially evaluating multiple candidate submodels. Additionally, based on different objective functions, we introduce two variants denoted as KFDB^{MSE} and KFDB^{ACC}.

- 3) We conduct extensive experiments to validate the superiority of KFDB on a collection of 60 UCI benchmark datasets. We also thoroughly analyze some key characteristics of KFDB by structural analysis, large datasets analysis, high-dimensional datasets analysis, and running time analysis.

The rest of this article is organized as follows. Section II reviews some existing structure-oriented improvements of NB. Section III proposes our KFDB. Section IV presents the experimental setup and results. Section V concludes this article and outlines the main directions for future work.

II. RELATED WORK

Structure-oriented improvements relax the attribute conditional independence assumption in NB by introducing directed arcs to explicitly represent dependencies between attributes. Generally speaking, structure-oriented improvements can be categorized into ODEs and KDEs.

A. One-Dependence Estimators

ODEs are the approaches that allow each attribute to be conditioned on the class and, at most, one other attribute. Different from BNC, given a test instance $\mathbf{x}$ represented by an attribute value vector $\langle a_1, a_2, \dots, a_m \rangle$, ODE utilizes (2) to predict its class label

$$c(\mathbf{x}) = \arg \max_{c \in C} P(c) \prod_{j=1}^m P(a_j | a_{jp}, c) \quad (2)$$

where a_j is the value of the j th attribute A_j of $\mathbf{x}$; a_{jp} is the value of its parent attribute A_{jp} ; and $P(c)$ and $P(a_j | a_{jp}, c)$ are the prior probability of c and the conditional probability of a_j given its parent a_{jp} and the class c , respectively, which can be estimated by the following equations:

$$P(c) = \frac{\sum_{i=1}^n \delta(c_i, c) + \frac{1}{q}}{n + 1} \quad (3)$$

$$P(a_j | a_{jp}, c) = \frac{\sum_{i=1}^n \delta(a_{ij}, a_j) \delta(a_{ijp}, a_{jp}) \delta(c_i, c) + \frac{1}{n_j}}{\sum_{i=1}^n \delta(a_{ijp}, a_{jp}) \delta(c_i, c) + 1} \quad (4)$$

where n is the number of training instances; q is the number of classes; c_i is the class label of the i th training instance; a_{ij} is the j th attribute value of the i th training instance; a_{ijp} is the value of A_{jp} of the i th training instance; n_j is the number of values for the j th attribute; and $\delta(\bullet)$ is a binary function, which is one if its two parameters are identical and zero otherwise.

To find the parent node of each attribute, inspired by the approach of Chow and Liu [25], Friedman et al. [10] proposed tree-augmented NB (TAN), in which the attribute dependencies are depicted using a tree-like structure. Specifically, TAN first constructs a maximum weighted spanning tree, where the weight of each arc is determined by conditional mutual information. Subsequently, a single attribute node is selected as the root node of this tree, converting the undirected spanning tree into a directed one. Unlike TAN, Keogh and Pazzani [26] proposed super-parent tree augmented NB (SP-TAN) based on the wrapper approach. SP-TAN employs a greedy heuristic search to learn the structure of TAN, always adding the arc that maximizes classification ACC at each iteration, until no further

addition of any possible arc can result in an improvement in classification ACC. Recently, Wang et al. [20] extended the one-dependence maximum weighted spanning tree to the high-dependence structure, and proposed scaled-up tree augmented NB (STAN). STAN first uses entropy-based scoring criteria to measure the fitness of the learned Bayesian network structure, and then, a heuristic search strategy is applied to identify high-dependence relationships. Finally, ensemble learning is introduced to further improve model robustness.

However, both TAN and its improvements require structure learning, which inevitably incur significant computational cost. To avoid structure learning, Webb et al. [11] proposed an improved model called averaged ODEs (AODE). AODE regards each attribute in turn as the parent node of all the other attributes and thus constructs multiple super-parent ODEs. Then, the final predicted results are generated by directly averaging the class membership probabilities of all the qualified super-parent ODEs. In AODE, each ODE is assigned the same weight, which means that all attributes are considered to have equal importance. However, in real-world applications, the importance of each ODE usually differs. To address this issue, Jiang et al. [27] proposed an improved model to assign different weights for different ODEs, which is called weighted AODEs (WAODE). In WAODE, the weights can be learned through four different methods, including mutual information, classification ACC, conditional log likelihood, and area under the receiver operating characteristic curve. All the models mentioned above are stable models. To conduct randomness into ODE, Jiang [28] further proposed random ODEs (RODE). In RODE, the parent node of each attribute is randomly selected from $\lfloor \log_2 m \rfloor$ attributes with the maximal conditional mutual information, where m is the total number of attributes.

B. K-Dependence Estimators

Different from ODE, KDE allows each attribute to depend on, at most, K other attributes apart from the class. Given a test instance $\mathbf{x}$ represented by an attribute value vector $\langle a_1, a_2, \dots, a_m \rangle$, KDE utilizes (5) to predict the class label of the test instance

$$c(\mathbf{x}) = \arg \max_{c \in C} P(c) \prod_{j=1}^m P(a_j | \pi_{jp}, c) \quad (5)$$

where π_{jp} is the parent set of attribute A_j . KDE still utilizes (3) to estimate the prior probability of the class c , but it utilizes (6) to estimate the conditional probability of a_j given the class c

$$P(a_j | \pi_{jp}, c) = \frac{\sum_{i=1}^n \delta(a_{ij}, a_j) \delta(\pi_{ijp}, \pi_{jp}) \delta(c_i, c) + \frac{1}{n_j}}{\sum_{i=1}^n \delta(\pi_{ijp}, \pi_{jp}) \delta(c_i, c) + 1} \quad (6)$$

where π_{ijp} is the attribute value of π_{jp} of the i th training instance.

The KDB [9] classifier is an exemplar of KDE. In KDB, the attributes are initially ranked in descending order based on their mutual information with the class. Then, the ordered attributes are incorporated into the KDB model sequentially. For each attribute, its parents are determined by choosing other attributes in the model that have the highest conditional mutual

information with it. Notably, the first K attributes added to the model will have fewer than K parents, while the remaining attributes will each have exactly K parents.

How to improve the structure of KDB and further enhance its classification performance is one of the most crucial issues. To this end, Rubio and Gámez [29] proposed a flexible KDB classifier, which employs a metaheuristic search procedure to find attribute order and K values simultaneously. Specifically, it treats the attribute order and parent nodes for each attribute as a set of vectors, and then, it utilizes the node histogram-based sampling algorithm to obtain new permutation. Finally, when the new permutation has been generated, the population is updated using a random-selection steady state scheme. Furthermore, Martínez et al. [22] proposed a selective KDB (SKDB) classifier to select the optimal attribute number and K value simultaneously by leave-one-out cross-validation (LOOCV), achieving a good tradeoff between structural complexity and classification performance. In addition, Liu et al. [30] presented an improved model called minimal-redundancy-maximal-relevance-based KDB with discriminative model selection. It first utilizes the minimal-redundancy-maximal-relevance criterion to sort attributes, and then, the discriminative model selection strategy is utilized to remove redundant attributes. Recently, Zhao et al. [31] proposed a dual learning-based flexible SKDB network (DL-FSKDB) classifier, which employs eager learning to construct the initial KDB model and then incorporates lazy learning for personalized fine-tuning and optimization.

In addition to a single KDE model, several ensemble learning methods have been proposed in the past few years. For example, Duan and Wang [32] introduced the K -dependence forest (KDF) classifier. In KDF, the number of base classifiers is equal to the number of attributes in the dataset, with each base classifier starting with a different attribute. Additionally, when determining the order of attributes, KDF takes into account not only the mutual information between attributes and class labels but also the conditional mutual information among attributes. Different from KDF, Zhang et al. [23] proposed another ensemble model called ensemble scalable KDB. Specifically, it constructs multiple stochastic models by varying the order of attributes and changing the discretization process for numerical attributes. In addition, an improved smoothing technique based on the hierarchical Dirichlet process is employed to further enhance the ACC of probability estimation in the model.

III. KFDB CLASSIFIERS

A. Motivation

Just as we discussed above, KDB enhances the classification performance by allowing each attribute to be conditioned on the class and, at most, K other attributes. However, the value of K , a crucial hyperparameter in KDB, typically needs to be pre-set by the user before model training. Unfortunately, obtaining sufficient prior knowledge to determine the most suitable K value for complex real-world applications can be challenging. To ensure the model can capture the dependencies between attributes as fully as possible, one common solution is to set

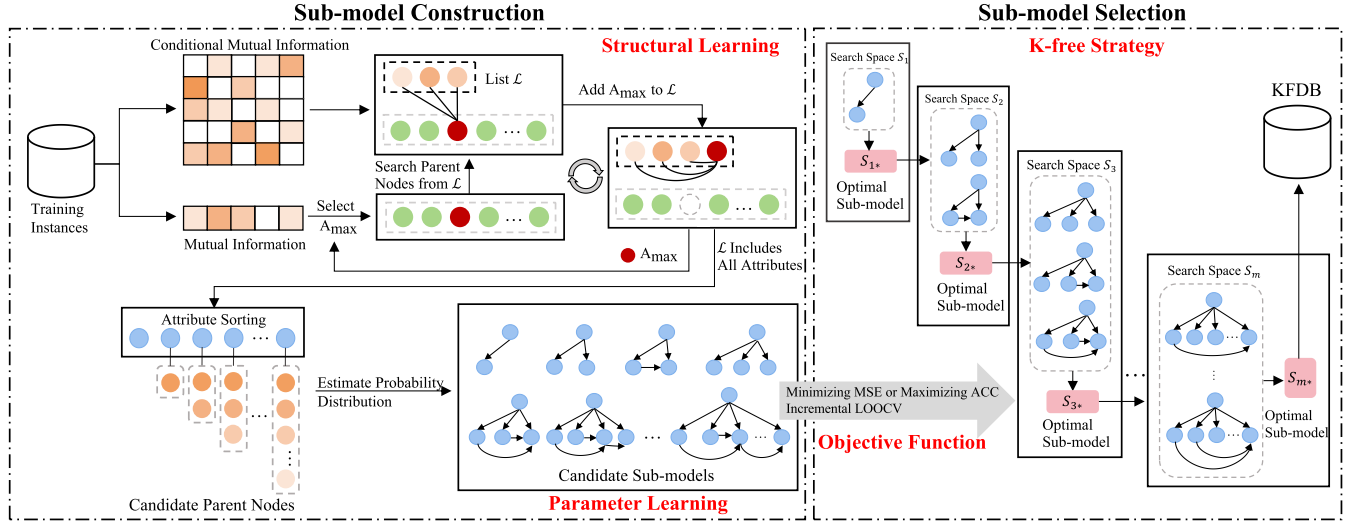


Fig. 1. Overall framework of KFDB.

a relatively large K value. However, as K becomes larger, the structural complexity of the KDB model grows significantly, which not only raises computational costs but also inevitably introduces a risk of overfitting. Alternatively, model selection techniques such as the Bayesian information criterion (BIC) [33] or minimum description length (MDL) [34] can also be employed to identify the most appropriate K value from a range of candidate models. Nevertheless, when K is given, the structure of KDB becomes immutable, meaning that all attribute dependencies are predetermined, which dramatically restricts the expressive power of the model. In this study, we argue that these two characteristics substantially hinder the performance of classical KDB. To overcome these challenges, we propose KFDB classifiers, which employ a greedy search strategy to adaptively learn the number of parent nodes for each attribute by sequentially evaluating multiple candidate submodels within a constrained search space. Moreover, to identify the most suitable network structure, we either minimize the MSE or maximize the classification ACC and thus propose two versions denoted as KFDB^{MSE} and KFDB^{ACC} , respectively.

B. Proposed Model

The overall framework of KFDB is illustrated in Fig. 1. As shown in the figure, KFDB belongs to a two-stage model. In the first stage, we greedily construct multiple candidate submodels based on a pre-set K value. In the second stage, we sequentially evaluate these submodels and adaptively learn the number of parent nodes for each attribute by minimizing the MSE or maximizing the classification ACC. Their detailed processes are introduced as follows.

1) *Stage 1 (Submodel Construction)*: Before describing the submodel construction processes in detail, we first formally introduce the definition of submodel in KFDB.

Definition 1 (Submodel): A submodel in KFDB refers to a specific Bayesian network structure that encompasses a subset of attributes along with their corresponding parent nodes. The submodel serves as the fundamental unit in the KFDB

construction process, facilitating the progressive learning and selection of the parent nodes for each attribute.

The submodel construction stage contains two modules: structural learning and parameter learning. Specifically, the structural learning module focuses on determining the structure of the Bayesian network, where we explore multiple candidate submodels and learn the possible parent node set for each attribute based on a pre-set K value. The parameter learning module focuses on estimating the probability distribution for the learned Bayesian network structure. In this module, we store the probability distribution of each attribute under different numbers of parent nodes, providing the basis for the subsequent submodel selection stage. In the following, we will present a detailed description of these two modules.

a) *Structural learning*: In the structural learning module, we first calculate the mutual information $MI(A_r; C)$ between each attribute and the class label, and the conditional mutual information $CM I(A_r; A_s | C)$ for each pair of attributes. We introduce these two definitions as follows.

Definition 2 (Mutual Information): Let A_r and C be two variables, and the mutual information between A_r and C is defined by (7), measuring their dependence degree

$$MI(A_r; C) = \sum_{a_r \in A_r} \sum_{c \in C} P(a_r, c) \log \frac{P(a_r, c)}{P(a_r)P(c)}. \quad (7)$$

Definition 3 (Conditional Mutual Information): Let A_r , A_s , and C be three variables, and the conditional mutual information between A_r and A_s given C is calculated by (8), measuring the dependence degree between A_r and A_s while conditioning on C

$$CM I(A_r; A_s | C) = \sum_{a_r \in A_r} \sum_{a_s \in A_s} \sum_{c \in C} P(a_r, a_s, c) \log \frac{P(a_r, a_s | c)}{P(a_r | c)P(a_s | c)}. \quad (8)$$

Following this, we initialize an empty list $\mathcal{L}$ to represent the used attributes, and construct the Bayesian network structure $\mathcal{G}$ with a single class node C . After that, we iteratively construct

the Bayesian network until all attributes are incorporated into the list $\mathcal{L}$. For the j th iteration, we first select an attribute A_{max}^j that is not yet in $\mathcal{L}$ and has the highest mutual information with the class label, which is described by

$$A_{max}^j = \arg \max_{A_r \notin \mathcal{L}} MI(A_r; C). \quad (9)$$

Then, this attribute is added as a node to $\mathcal{G}$, and a directed arc is established from the class C to the attribute A_{max}^j . Subsequently, we employ (10) to select b_j different attributes from the list $\mathcal{L}$ as its candidate parent nodes, which have the highest conditional mutual information with the attribute A_{max}^j

$$\pi_{jp} = \arg \max_{A_s \in \mathcal{L}} CMI(A_{max}^j; A_s | C), \quad \text{s.t.} \quad |\pi_{jp}| = b_j \quad (10)$$

where π_{jp} is the candidate parent node set of A_{max}^j , and b_j is the minimum of the current size of $\mathcal{L}$ and the pre-set value of K , i.e., $b_j = \min(|\mathcal{L}|, K)$. Ultimately, we add a directed arc from each candidate parent node in π_{jp} to A_{max}^j in $\mathcal{G}$, and insert the attribute A_{max}^j into $\mathcal{L}$. Note that the first K attributes added to the model will have fewer than K candidate parent nodes, while the remaining attributes will each have exactly K candidate parent nodes.

b) Parameter learning: After the structural learning module is completed, we proceed to obtain multiple candidate submodels $\mathcal{M}$ by estimating the needed parameters associated with the learned structure. Specifically, we begin by creating the data structure for the conditional probability tables Θ based on the network structure $\mathcal{G}$, where the array structure of each attribute is defined according to the number of potential parent nodes, and each element in the array is initialized to zero. Subsequently, the training instances D are employed to estimate each parameter value in the conditional probability tables Θ . For the attribute A_{max}^j conditioned on v parent nodes, the parameter value θ_j^v in the conditional probability table, i.e., the corresponding conditional probability of a_j with v parents, is estimated as follows:

$$\theta_j^v = \frac{\sum_{i=1}^n \delta(a_{ij}, a_j) \delta(\pi_{ijp}^v, \pi_{jp}^v) \delta(c_i, c) + \frac{1}{n_j}}{\sum_{i=1}^n \delta(\pi_{ijp}^v, \pi_{jp}^v) \delta(c_i, c) + 1} \quad (11)$$

where π_{jp}^v denotes the parent node set of A_{max}^j with a size of v , i.e., a_j containing v parent nodes, and π_{ijp}^v is the attribute value of π_{jp}^v of the i th training instance.

Therefore, multiple candidate submodels $\mathcal{M}$ are constructed by combining different attributes and assigning varying numbers of parent nodes to them. The complete submodel construction algorithm is summarized in Algorithm 1.

2) Stage 2 (Submodel Selection): In the submodel selection stage, we sequentially evaluate these constructed candidate submodels and select the one that achieves local optimal performance. This stage primarily addresses two challenges:

- 1) how to use the K-free strategy to determine the number of parent nodes for each attribute.
- 2) how to define the objective function and evaluate each candidate submodel.

We will provide a detailed discussion of these two challenges as follows.

Algorithm 1 Submodel Construction (D, K)

Input: a training dataset D , the maximum number of parent node K .

Output: multiple candidate sub-models $\mathcal{M}$.

- 1: Calculate $MI(A_r; C)$ from D for each attribute by Eq. (7);
 - 2: Calculate $CMI(A_r; A_s | C)$ from D for each pair of attributes A_r and $A_s (r \neq s)$ by Eq. (8);
 - 3: Initialize an empty list $\mathcal{L}$ to represent the used attributes;
 - 4: Construct the Bayesian network structure $\mathcal{G}$ with a single class node C ;
 - // **Structural learning**
 - 5: **for** $j = 1$ to m **do**
 - 6: Select an attribute A_{max}^j , which is not yet in $\mathcal{L}$, and has the highest value $MI(A_{max}^j; C)$ by Eq. (9);
 - 7: Add a node to $\mathcal{G}$ representing A_{max}^j ;
 - 8: Add an arc from C to A_{max}^j in $\mathcal{G}$;
 - 9: Select $b_j = \min(|\mathcal{L}|, K)$ different attributes from $\mathcal{L}$ as the candidate parent nodes of A_{max}^j , which have the highest value $CMI(A_{max}^j; A_s | C)$ with A_{max}^j by Eq. (10);
 - 10: Add the directed arcs from each candidate parent node in π_{jp} to A_{max}^j in $\mathcal{G}$;
 - 11: Add A_{max}^j to $\mathcal{L}$;
 - 12: **end for**
 - // **Parameter learning**
 - 13: Create the data structure for conditional probability tables Θ based on $\mathcal{G}$;
 - 14: Estimate the values of each parameter in Θ from D by Eq. (11);
 - 15: Construct multiple candidate sub-models $\mathcal{M}$ by combining different attributes and assigning varying numbers of parent nodes to them;
 - 16: **return** $\mathcal{M}$
-

a) K-free strategy: To address the first challenge, in KFDB, we propose the K-free strategy, i.e., a greedy search strategy to sequentially determine the number of parent nodes for each attribute, iteratively adding each attribute with its dependency arcs into the previously selected submodels. In other words, we iteratively select the current optimal submodel under the assumption that previously selected submodels are fixed, continuing this process until all attributes have been incorporated. Specifically, we begin by constructing m empty search spaces, denoted as $\langle \mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m \rangle$. Then, we start with the first attribute, where its search space $\mathcal{S}_1$ contains only one submodel $\mathcal{S}_{10}$: the attribute A_1 with no parent nodes. Next, we traverse and select one optimal submodel for each attribute iteratively. For the j th attribute A_j , we search in its search space $\mathcal{S}_j$ based on previously selected submodel $\mathcal{S}_{(j-1)*}$ and select current optimal submodel $\mathcal{S}_{j*}$ within its search space $\mathcal{S}_j$. Here, it is important to emphasize that KFDB fully exploits a key property of BNC in posterior probability estimation; each submodel is nested within another submodel that includes one additional attribute. In other words, the posterior probability estimated by the smaller submodel can be fully reused by the larger one. When we evaluate the probability terms of the j th attribute submodel within its search space $\mathcal{S}_j$, the required posterior probabilities are computed by multiplying the posterior probability obtained from the previously selected submodel $\mathcal{S}_{(j-1)*}$ with the conditional probabilities of the j th attribute. Therefore, compared to KDB, KFDB requires only one extra pass over the training data, which does not incur significant computational overhead and remains highly efficient.

To further illustrate the proposed novel K-free strategy, we present a toy example. Suppose that in the submodel construction stage, the maximum value of K is set to 2, and the attributes, ordered by mutual information, are

$\langle A_1, A_2, A_3, A_4 \rangle$. In this scenario, A_1 has no parent node. When searching for the parent nodes for A_2 , all the necessary conditional probabilities involving A_1 have already been calculated. Therefore, we only need to evaluate the performance of the candidate submodels $\mathcal{S}_{20}$ and $\mathcal{S}_{21}$, corresponding to the cases where A_2 has 0 and 1 parent nodes, respectively. Assume that submodel $\mathcal{S}_{21}$ performs better than $\mathcal{S}_{20}$, and $\mathcal{S}_{21}$ will be selected as current optimal submodel. Next, we proceed to search for the parent nodes for A_3 , evaluating the performance of the candidate submodels $\mathcal{S}_{30}$, $\mathcal{S}_{31}$, and $\mathcal{S}_{32}$, and adding the latest optimal one among them to the previously selected submodel $\mathcal{S}_{21}$. This procedure is repeated iteratively until all attributes have been considered, ensuring that each subsequently selected submodel in $\mathcal{S}_j$ is constructed upon the previously selected submodel $\mathcal{S}_{(j-1)*}$.

b) Objective function: To address the second challenge, we employ MSE and classification ACC as objective functions to evaluate the candidate submodels, resulting in two versions, which we refer to as KFDB^{MSE} and KFDB^{ACC}. Specifically, MSE calculates mean squared errors of all training instances between their predicted probability and the true probability of the actual class, while ACC measures the proportion of correctly classified instances relative to the total number of instances. For the v th candidate submodel of the j th attribute $\mathcal{S}_{jv}$, the MSE function is formally defined in the following equation:

$$MSE_{jv} = \frac{1}{n} \sum_{i=1}^n \left(P(c_i | y_i) - \hat{P}_{jv}(c_i | y_i) \right)^2 \quad (12)$$

where n is the number of instances, y_i is the i th training instance, c_i is the true class label of the i th training instance, and $P(c_i | y_i)$ denotes the true probability of y_i belonging to c_i . Since the actual class of y_i is known to be c_i , we set $P(c_i | y_i) = 1.0$. Moreover, $\hat{P}_{jv}(c_i | y_i)$ denotes the predicted probability of assigning y_i to class c_i under submodel $\mathcal{S}_{jv}$, which is computed as follows:

$$\hat{P}_{jv}(c_i | y_i) = \frac{P(c_i) \prod_{j'=1}^j P_{jv}(a_{j'} | \pi_{j'p}^v, c_i)}{\sum_{c \in C} P(c) \prod_{j'=1}^j P_{jv}(a_{j'} | \pi_{j'p}^v, c)} \quad (13)$$

where $P(c_i)$ is the prior probability of class c_i , $P_{jv}(a_{j'} | \pi_{j'p}^v, c_i)$ is the conditional probability of $a_{j'}$ given the class c_i and parent node set $\pi_{j'p}^v$ by $\mathcal{S}_{jv}$, j' is an index ranging from 1 to j , and C is the collection of all possible class labels c .

Similarly, we define the ACC objective function as follows:

$$ACC_{jv} = \frac{1}{n} \sum_{i=1}^n \delta(c_i, \hat{c}_i^{jv}) \quad (14)$$

where

$$\delta(c_i, \hat{c}_i^{jv}) = \begin{cases} 1, & c_i = \hat{c}_i^{jv} \\ 0, & c_i \neq \hat{c}_i^{jv} \end{cases} \quad (15)$$

in which c_i denotes the true class of the i th training instance and $\hat{c}_i^{jv}$ denotes the predicted class of the i th training instance under submodel $\mathcal{S}_{jv}$ obtained using the maximum a posteriori (MAP) rule. Specifically, the posterior probability of each class is estimated under the network structure of $\mathcal{S}_{jv}$, and the class with the highest posterior probability is assigned as $\hat{c}_i^{jv}$.

Algorithm 2 Submodel Selection ($D, K, \mathcal{M}$)

Input: a training dataset D , the maximum number of parent K , all the submodels $\mathcal{M}$.

Output: KFDB model $\mathcal{A}$.

```

// K-free strategy
1: for each attribute  $A_j (j = 1, 2, \dots, m)$  do
2:   Initialize an empty search space  $\mathcal{S}_j$  for the attribute  $A_j$ ;
3:   if  $j = 1$  then
4:     Select a sub-model from  $\mathcal{M}$  into  $\mathcal{S}_1$ , including the attribute  $A_1$ 
       with no parent nodes;
5:   else
6:     Select  $b_j + 1$  sub-models from  $\mathcal{M}$  into  $\mathcal{S}_j$  based on the selected
       sub-model  $\mathcal{S}_{(j-1)*}$ ;
// Objective function
7:   for each sub-model  $\mathcal{S}_{jv} (v = 0, 1, \dots, b_j)$  from  $\mathcal{S}_j$  do
8:     for each training instance  $y_i (i = 1, 2, \dots, n)$  from  $D$  do
9:       Estimate the posterior probability  $\hat{P}_{jv}(c_i | y_i)$  using  $\mathcal{S}_{jv}$  by
       Eq. (13);
10:    end for
11:    Calculate the  $MSE_{jv}$  or  $ACC_{jv}$  of  $\mathcal{S}_{jv}$  by Eq. (12) or Eq. (14);
12:  end for
13:  Select a sub-model  $\mathcal{S}_{j*}$  with the minimum  $MSE$  or the maximum
     ACC by Eq. (16) or Eq. (17);
14: end if
15: end for

```

Therefore, the optimal submodel $\mathcal{S}_{j*}$ for the j th attribute is determined by evaluating all its candidate submodels and selecting the one that yields the best objective function value, as defined in the following:

$$\mathcal{S}_{j*} = \arg \min_{v \in \{0, \dots, b_j\}} \frac{1}{n} \sum_{i=1}^n \left(P(c_i | y_i) - \hat{P}_{jv}(c_i | y_i) \right)^2 \quad (16)$$

$$\mathcal{S}_{j*} = \arg \max_{v \in \{0, \dots, b_j\}} \frac{1}{n} \sum_{i=1}^n \delta(c_i, \hat{c}_i^{jv}) \quad (17)$$

where v denotes the v th candidate submodel within the search space of attribute A_j , i.e., assigning v parent nodes to the given attribute A_j , with the range from 0 to b_j . Here, b_j indicates the total number of submodels in the j th search space.

In addition, to provide an unbiased estimation of the generalization error, we evaluate the candidate submodels using LOOCV. In LOOCV, each instance in the dataset is sequentially treated as the validation instance, while the remaining instances constitute the training set. This strategy enables a comprehensive evaluation of model performance by predicting each instance in the dataset. However, LOOCV usually leads to severe computational costs. To mitigate this, we adopt an incremental LOOCV strategy to evaluate all candidate submodels, which is particularly well-suited for BNC. Specifically, during the submodel construction stage, we first traverse all the training instances to count the frequencies of each class and each pair of attribute values. Then, when evaluating a candidate submodel, we simply adjust the frequency counts by temporarily excluding the validation instance, rather than rebuilding the entire BNC, thereby substantially reducing the computational overhead. The complete submodel selection algorithm is summarized in Algorithm 2.

Please note that although KFDB still requires a pre-set maximum value of K for submodel construction, its role and interpretation within KFDB differ fundamentally from those in KDB. Specifically, in KDB, except for the first K attributes, all other attributes are constrained to have exactly

TABLE I
CLASSIFICATION ACC COMPARISONS FOR KFDB^{MSE} VERSUS KDB AT DIFFERENT VALUES OF K

Dataset	K=1		K=2		K=3		K=4		K=5	
	KFDB ^{MSE}	KDB	KFDB ^{MSE}	KDB	KFDB ^{MSE}	KDB	KFDB ^{MSE}	KDB	KFDB ^{MSE}	KDB
anneal	98.83	98.61	98.94	98.44	98.83	98.22	98.78	97.89	98.89	97.66
anneal.ORIG	91.76	91.93	91.59	92.04	92.20	91.59	91.93	90.37	91.42	89.42
artificial	57.90	57.90	62.74	62.74	67.14	67.14	68.47	68.47	68.72	68.79
audiology	71.69	70.78	71.67	66.36	71.68	65.47	72.57	61.48	71.67	55.05
autos	77.80	76.83	80.24	78.78	79.27	78.78	79.27	78.54	79.02	78.05
balance-scale	90.16	85.04	90.16	75.12	90.16	69.84	90.16	69.84	90.16	69.84
breast-cancer	70.11	68.38	69.06	63.64	67.49	64.34	67.32	63.99	67.49	63.29
breast-w	97.07	95.07	97.00	94.28	96.92	94.21	96.92	94.13	96.92	94.21
car	94.21	94.21	95.02	94.97	96.01	95.60	95.92	94.73	95.92	94.71
cardiotoc	90.43	89.91	90.71	90.54	90.85	90.31	90.52	89.91	90.80	89.18
climate	90.09	90.09	88.52	70.93	87.96	75.46	87.96	76.02	87.96	76.11
colic	78.66	77.05	78.12	76.21	78.26	71.73	78.54	72.00	78.12	72.28
colic.ORIG	75.54	77.30	75.54	77.30	75.54	77.30	75.68	77.44	75.68	77.44
connectionist	82.29	82.67	84.46	83.70	83.89	82.37	83.99	82.19	83.99	82.47
credit-a	84.13	82.10	84.13	76.74	84.35	77.32	84.42	77.54	84.42	77.97
credit-g	74.20	72.15	73.70	70.90	73.25	69.65	73.10	65.95	73.10	65.20
cylinder-bands	79.63	77.41	79.63	77.41	79.35	77.50	79.44	77.31	79.44	77.41
diabetes	74.94	74.80	74.94	68.36	74.94	66.60	74.94	67.77	74.94	67.19
ecoli	82.88	76.78	82.44	71.01	82.44	70.11	82.44	70.25	82.44	70.10
energy-y1	68.16	65.82	69.14	66.93	69.14	66.47	69.08	69.01	69.08	69.20
energy-y2	53.84	51.04	53.52	50.78	54.23	53.71	54.36	55.28	54.23	55.93
glass	57.96	57.71	57.02	57.72	57.25	60.99	57.96	60.52	58.42	58.89
hayes-roth	75.31	69.69	72.50	68.44	72.50	68.44	72.50	68.44	72.50	68.44
heart-c	81.84	75.24	80.69	70.80	81.02	70.30	81.02	69.96	81.02	69.80
heart-h	81.29	76.03	80.95	74.99	80.60	73.46	80.60	72.95	80.60	72.61
heart-statlog	79.44	76.30	79.07	72.04	78.89	70.56	78.89	68.89	78.89	68.52
hepatitis	84.84	79.68	82.58	79.68	81.29	79.68	81.29	77.42	81.29	78.06
hypothyroid	93.25	92.93	93.21	92.43	93.20	92.47	93.12	92.42	93.15	92.06
ionosphere	93.16	92.74	92.88	93.30	92.45	92.17	92.73	90.61	93.44	91.32
iris	94.67	94.33	94.67	95.00	94.33	95.00	94.33	95.00	94.33	95.00
kr-vs-kp	95.35	94.41	95.85	95.64	96.40	95.62	97.00	96.23	97.48	96.87
labor	90.53	88.71	93.03	84.39	93.86	86.14	93.86	86.14	93.86	86.14
letter	83.62	83.62	88.10	87.96	88.90	88.92	88.78	88.43	88.80	87.79
libras	74.31	75.56	74.72	75.14	75.00	75.14	74.72	75.83	74.58	75.83
lymph	84.43	82.74	83.07	77.00	82.76	74.99	81.38	78.37	80.71	76.99
mfeat-f	78.88	78.20	78.20	71.50	78.25	67.47	77.55	65.68	77.68	65.13
monks	100.00	100.00	100.00	100.00	100.00	98.83	100.00	94.24	100.00	93.08
mushroom	99.93	99.87	100.00	100.00	100.00	99.98	100.00	100.00	100.00	100.00
newthyroid	93.26	92.79	93.02	91.16	92.79	91.40	92.79	91.40	92.79	91.40
optdigits	94.29	94.35	94.55	93.04	94.24	91.03	94.31	89.19	94.10	87.79
page-blocks	93.45	93.17	93.71	93.69	93.77	93.66	93.77	93.59	93.77	93.61
parkinsons	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26
pendigits	96.56	96.52	97.78	97.75	97.80	97.35	97.84	96.68	97.84	96.22
primary	45.14	45.28	45.43	43.81	45.14	41.30	44.10	39.67	43.95	38.94
qar	84.45	83.32	84.27	84.17	84.12	82.51	83.84	81.37	83.27	80.19
robot-24	87.77	87.70	90.69	90.79	91.37	91.38	91.46	91.77	91.88	91.95
segment	94.61	94.65	94.83	94.87	95.00	94.81	95.00	94.74	94.72	94.35
sick	97.69	97.80	97.79	97.79	98.05	97.81	98.03	97.89	98.01	97.93
sonar	72.56	72.82	77.39	75.21	77.87	74.03	77.64	72.60	77.15	72.12
soybean	94.73	94.65	94.80	93.26	94.43	92.82	94.14	92.46	94.43	92.38
spectrometer	50.28	50.95	51.88	49.53	51.88	47.46	51.89	46.24	49.53	45.39
splice	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49
steel-plates	95.72	91.14	96.06	90.06	96.29	87.30	96.26	87.17	95.96	86.97
texture	94.37	94.30	96.07	96.14	96.72	96.80	96.76	96.71	96.80	96.64
thyroid	93.88	93.69	93.86	93.63	93.84	93.36	93.81	93.24	93.81	92.93
vehicle	73.58	73.46	73.17	71.22	72.99	70.80	72.40	69.03	72.40	69.44
vote	94.14	94.14	94.48	94.60	95.17	95.17	95.06	94.14	95.06	94.37
vowel	85.15	85.10	83.99	84.24	83.59	83.13	83.79	83.03	83.64	83.23
waveform	82.03	79.35	81.96	74.36	81.96	66.30	81.96	58.32	81.96	53.89
zoo	95.60	95.60	95.60	95.60	95.60	94.10	95.60	94.10	95.10	79.10
Average	83.80	82.67	84.08	81.40	84.15	80.87	84.13	80.34	84.05	79.96
Better	41	10	41	11	49	7	50	6	50	7
W/T/L	-	22/36/2	-	28/31/1	-	30/29/1	-	30/29/1	-	33/26/1

K parent nodes, whereas KFDB allows each attribute to learn an adaptive number of parent nodes, ranging from 0 to K. Meanwhile, leveraging the pre-set K, the proposed K-free strategy substantially enhances the flexibility of KDB and provides a practical tradeoff between computational feasibility and strong classification performance.

C. Time Complexity Analysis

KFDB is a structure-oriented improved BNC, wherein the primary computational cost arises from the process of learning the Bayesian network structures. Based on Algorithm 1, in the submodel construction stage, we first need to calculate the mutual information and the conditional mutual information

for each pair of attributes, which takes the time complexity of $O(qmt)$ and $O(qm^2t^2)$, respectively, where q is the number of classes, m is the number of attributes, t is the maximum number of discrete values any attribute can take. Next, we sort the attributes and learn the potential parent nodes of each attribute, which takes time complexity of $O(m\log m)$ and $O(m^2\log m)$, respectively. Based on Algorithm 2, in the submodel selection stage, we sequentially evaluate the candidate submodels using incremental LOOCV to select the optimal one, which takes a time complexity of $O(nqm^2K^2)$, where n is the number of training instances. Owing to the nested formulation of posterior probability estimation in BNC and the use of the incremental LOOCV strategy, the submodel selection stage requires only a single pass over

TABLE II
CLASSIFICATION ACC COMPARISONS FOR KFDB^{ACC} VERSUS KDB AT DIFFERENT VALUES OF K

Dataset	K=1		K=2		K=3		K=4		K=5	
	KFDB ^{ACC}	KDB	KFDB ^{ACC}	KDB	KFDB ^{ACC}	KDB	KFDB ^{ACC}	KDB	KFDB ^{ACC}	KDB
anneal	98.39	98.61	98.94	98.44	99.28	98.22	99.05	97.89	99.05	97.66
anneal.ORIG	91.48	91.93	91.54	92.04	91.70	91.59	91.93	90.37	91.82	89.42
artificial	57.69	57.90	62.59	62.74	66.73	67.14	68.22	68.47	68.58	68.79
audiology	74.34	70.78	71.68	66.36	73.47	65.47	73.67	61.48	72.57	55.05
autos	76.83	76.83	78.54	78.78	78.54	78.78	78.29	78.54	78.29	78.05
balance-scale	90.32	85.04	90.32	75.12	90.32	69.84	90.32	69.84	90.32	69.84
breast-cancer	69.42	68.38	67.84	63.64	67.67	64.34	68.36	63.99	68.36	63.29
breast-w	97.14	95.07	97.14	94.28	97.21	94.21	97.21	94.13	97.21	94.21
car	93.64	94.21	94.73	94.97	94.99	95.60	94.76	94.73	94.76	94.71
cardiotoc	89.86	89.91	90.97	90.54	90.62	90.31	90.50	89.91	90.17	89.18
climate	88.80	90.09	86.67	70.93	86.48	75.46	86.48	76.02	86.48	76.11
colic	77.99	77.05	78.39	76.21	75.96	71.73	76.89	72.00	76.76	72.28
colic.ORIG	75.27	77.30	75.27	77.30	75.27	77.30	75.41	77.44	75.41	77.44
connectionist	83.33	82.67	83.42	83.70	83.70	82.37	83.23	82.19	83.23	82.47
credit-a	83.84	82.10	84.06	76.74	84.57	77.32	84.93	77.54	85.07	77.97
credit-g	72.00	72.15	71.25	70.90	71.60	69.65	72.45	65.95	71.75	65.20
cylinder-bands	81.02	77.41	80.93	77.41	80.93	77.50	80.93	77.31	80.93	77.41
diabetes	75.32	74.80	75.32	68.36	74.48	66.60	74.15	67.77	74.15	67.19
ecoli	81.25	76.78	81.10	71.01	81.10	70.11	81.10	70.25	81.10	70.10
energy-y1	65.88	65.82	67.12	66.93	67.25	66.47	67.71	69.01	67.71	69.20
energy-y2	52.48	51.04	52.48	50.78	53.52	53.71	53.52	55.28	53.52	55.93
glass	58.64	57.71	58.62	57.72	61.89	60.99	61.19	60.52	61.42	58.89
hayes-roth	80.94	69.69	78.75	68.44	78.75	68.44	78.75	68.44	78.75	68.44
heart-c	82.83	75.24	79.38	70.80	79.54	70.30	79.54	69.96	79.54	69.80
heart-h	80.95	76.03	81.12	74.99	80.28	73.46	80.11	72.95	80.11	72.61
heart-statlog	78.70	76.30	78.33	72.04	78.70	70.56	79.07	68.89	79.07	68.52
hepatitis	83.87	79.68	82.26	79.68	81.29	79.68	81.94	77.42	81.94	78.06
hypothyroid	93.08	92.93	92.97	92.43	92.97	92.47	93.00	92.42	92.99	92.06
ionosphere	92.45	92.74	91.73	93.30	92.31	92.17	92.59	90.61	92.17	91.32
iris	94.67	94.33	94.67	95.00	94.67	95.00	94.67	95.00	94.67	95.00
kr-vs-kp	91.86	94.41	92.69	95.64	94.02	95.62	94.24	96.23	93.95	96.87
labor	92.27	88.71	93.18	84.39	93.18	86.14	93.18	86.14	93.18	86.14
letter	83.60	83.62	88.01	87.96	88.77	88.92	88.70	88.43	88.47	87.79
libras	73.47	75.56	73.19	75.14	74.17	75.14	73.89	75.83	74.58	75.83
lymph	84.77	82.74	83.75	77.00	80.40	74.99	82.78	78.37	81.75	76.99
mfeat-f	78.55	78.20	77.72	71.50	76.38	67.47	76.47	65.68	76.75	65.13
monks	100.00	100.00	100.00	100.00	100.00	98.83	100.00	94.24	100.00	93.08
mushroom	99.83	99.87	99.98	100.00	99.99	99.98	99.99	100.00	100.00	100.00
newthyroid	93.95	92.79	93.72	91.16	93.72	91.40	93.72	91.40	93.72	91.40
optdigits	94.08	94.35	94.40	93.04	94.40	91.03	93.98	89.19	94.21	87.79
page-blocks	93.37	93.17	93.70	93.69	93.80	93.66	93.84	93.59	93.87	93.61
parkinsons	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26	80.26
pendigits	96.59	96.52	97.73	97.75	97.80	97.35	97.58	96.68	97.58	96.22
primary	44.55	45.28	45.43	43.81	44.09	41.30	43.06	39.67	43.50	38.94
qar	83.51	83.32	84.69	84.17	84.50	82.51	83.36	81.37	83.46	80.19
robot-24	87.16	87.70	90.40	90.79	91.23	91.38	91.69	91.77	91.72	91.95
segment	94.39	94.65	94.63	94.87	94.63	94.81	94.52	94.74	94.76	94.35
sick	97.73	97.80	97.83	97.79	97.81	97.81	97.85	97.89	97.77	97.93
sonar	72.58	72.82	73.52	75.21	75.46	74.03	75.95	72.60	75.95	72.12
soybean	94.29	94.65	94.73	93.26	93.77	92.82	94.21	92.46	94.14	92.38
spectrometer	51.04	50.95	51.32	49.53	51.32	47.46	50.66	46.24	50.57	45.39
splice	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49	95.49
steel-plates	95.70	91.14	95.03	90.06	95.29	87.30	95.34	87.17	95.03	86.97
texture	94.25	94.30	95.90	96.14	96.39	96.80	96.47	96.71	96.65	96.64
thyroid	93.53	93.69	93.56	93.63	93.47	93.36	93.35	93.24	93.41	92.93
vehicle	72.46	73.46	71.16	71.22	71.34	70.80	71.75	69.03	70.98	69.44
vote	93.22	94.14	94.37	94.60	94.48	95.17	95.52	94.14	95.06	94.37
vowel	84.24	85.10	83.74	84.24	84.04	83.13	83.94	83.03	83.84	83.23
waveform	82.05	79.35	81.73	74.36	81.73	66.30	81.73	58.32	81.73	53.89
zoo	95.60	95.60	95.60	95.60	95.60	94.10	95.60	94.10	95.60	94.10
Average	83.61	82.67	83.76	81.40	83.89	80.87	83.98	80.34	83.93	79.96
Better	30	25	36	20	44	13	45	13	48	9
W/T/L	-	12/44/4	-	21/36/3	-	25/30/5	-	33/22/5	-	36/21/3

the training data. Consequently, compared with the classical KDB, which requires two passes over the training data, the proposed KFDB necessitates only a single additional pass. In summary, the overall time complexity of the training stage is $O(qmt + qm^2t^2 + m\log m + m^2\log m + nqm^2K^2)$, which is slightly higher than the classical KDB and is still very efficient.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setting and Benchmark Data

In this section, we designed a series of experiments to evaluate the performance of our proposed KFDB. First, we compared KFDB with the classical KDB at the same K values to validate the effectiveness of the proposed K-free strategy. Then, we compared KFDB with several closely related

state-of-the-art models, including DL-FSKDB, KDF, KDB, STAN, TAN, WANBIA, NB, RF, and LR, to validate the superiority of KFDB in terms of classification ACC. In addition, we performed some in-depth discussions, including the structural analysis, large dataset analysis, and high-dimensional dataset analysis, to demonstrate some key characteristics of KFDB. In our experiments, we implemented KFDB, TAN, and NB and used the existing LR and RF models on the Waikato environment for knowledge analysis (WEKA) platform. The source code of DL-FSKDB, KDF, KDB, STAN, and WANBIA was kindly provided by the original authors. These models and their abbreviations are listed as follows.

- 1) $KFDB^{MSE}$: KFDB classifier with objective function of MSE.

- 2) $KFDB^{ACC}$: KFDB classifier with objective function of classification ACC.
- 3) $DL-FSKDB$: Dual learning-based flexible SKDB network classifier [31].
- 4) KDF : K-dependence forest classifier [32].
- 5) KDB : K-dependence Bayesian classifier [9].
- 6) $STAN$: Scaled-up tree-augmented NB [20].
- 7) TAN : Tree-augmented NB [10].
- 8) $WANBIA$: Attribute weighted NB with objective function of MSE [35].
- 9) NB : Naive Bayes [36].
- 10) RF : Random forest [37].
- 11) LR : Logistic regression [38].

Our experiments were conducted on a collection of 60 benchmark classification datasets from the UCI repository, which are the same as those of recent peer work [16] and represent a wide range of domains and data characteristics. In our experiments, all missing attribute values were replaced with modes for nominal attributes and the means for numerical attributes from the available data. Besides, since all models could only handle nominal attributes, numerical attributes were discretized into nominal ones by dividing them into ten equal-width bins. Finally, we manually removed four useless attributes: “Hospital Number” from the “colic.ORIG” dataset, “name” from the “parkinsons” dataset, “instance name” from the “splice” dataset, and “animal” from the “zoo” dataset. These attributes were discarded because their number of distinct values was nearly equal to the number of instances in the dataset, rendering them ineffective for classification. To ensure reproducibility of this article and allow other researchers to readily build upon our work, we have made the source code of KFDB publicly available on <https://github.com/zhanghuan1994/KFDB>.

B. Experimental Results and Analysis

All experimental results were obtained from two separate runs using stratified fivefold cross-validation. Tables I and II present the detailed classification ACC comparisons for $KFDB^{MSE}$ and $KFDB^{ACC}$ versus KDB, respectively. In each row, a field marked with • or ◦ indicates that our proposed models significantly outperform or underperform their competitors, based on a corrected paired two-tailed t-test at the $p = 0.05$ significance level. Additionally, we display the average classification ACC, the number of datasets on which the corresponding method achieved better results, and the *Win/Tie/Lose*(*W/T/L*) results on 60 benchmark datasets. Each entry’s *W/T/L* denotes that, compared to their corresponding competitors, our proposed $KFDB^{MSE}$ and $KFDB^{ACC}$ win on *W* datasets, tie on *T* datasets, and loss on *L* datasets.

Moreover, to conduct a thorough comparison between each pair of models, based on the ACC results in Tables I and II, we employed the knowledge extraction based on evolutionary learning (KEEL) data mining software tool to perform the Wilcoxon signed-rank test. The detailed comparison results are summarized in Tables III and IV. In these tables, the symbol • and ◦ indicate that the model in the column improves or degrades the model in the corresponding row. The lower

TABLE III
CLASSIFICATION ACC COMPARISONS OF THE WILCOXON TEST FOR $KFDB^{MSE}$ VERSUS KDB AT DIFFERENT VALUES OF K

	$K=1$ $KFDB^{MSE}$	KDB	$K=2$ $KFDB^{MSE}$	KDB	$K=3$ $KFDB^{MSE}$	KDB	$K=4$ $KFDB^{MSE}$	KDB	$K=5$ $KFDB^{MSE}$	KDB
$K=1$ $KFDB^{MSE}$		•		◦						
$K=2$ $KFDB^{MSE}$				•		◦				
$K=3$ $KFDB^{MSE}$						•		◦		
$K=4$ $KFDB^{MSE}$								•		◦
$K=5$ $KFDB^{MSE}$									•	◦

TABLE IV
CLASSIFICATION ACC COMPARISONS OF THE WILCOXON TEST FOR $KFDB^{ACC}$ VERSUS KDB AT DIFFERENT VALUES OF K

	$K=1$ $KFDB^{ACC}$	KDB	$K=2$ $KFDB^{ACC}$	KDB	$K=3$ $KFDB^{ACC}$	KDB	$K=4$ $KFDB^{ACC}$	KDB	$K=5$ $KFDB^{ACC}$	KDB
$K=1$ $KFDB^{ACC}$		•		◦						
$K=2$ $KFDB^{ACC}$				•		◦				
$K=3$ $KFDB^{ACC}$						•		◦		
$K=4$ $KFDB^{ACC}$								•		◦
$K=5$ $KFDB^{ACC}$									•	◦

diagonal represents a significance level of $\alpha = 0.05$, and the upper diagonal corresponds to a significance level of $\alpha = 0.1$.

Based on all these results, we conclude that regardless of the K value, our proposed $KFDB^{MSE}$ and $KFDB^{ACC}$ significantly outperform their corresponding KDB. Now, we summarize the highlights as follows.

- 1) *The Improvement in Averaged Classification ACC*: From Tables I and II, the averaged classification ACC of $KFDB^{MSE}$ on 60 datasets, with K values ranging from 1 to 5, is 83.80%, 84.08%, 84.15%, 84.13% and 84.05%. Meanwhile, the averaged result of $KFDB^{ACC}$ is 83.61%, 83.76%, 83.89%, 83.98%, and 83.93%. All of them are remarkably higher than the corresponding KDB with the same K values, whose classification accuracies are 82.67%, 81.40%, 80.87%, 80.34%, and 79.96%. Therefore, compared to KDB, KFDB consistently improves the classification ACC, regardless of the value of K, and whether the objective function is MSE or classification ACC.
- 2) *The Effectiveness in Preventing Overfitting and Increasing Model Expressiveness*: From the results in Tables I and II, it is evident that as K increases, the averaged classification ACC of KDB consistently decreases, while the averaged classification ACC of both $KFDB^{MSE}$ and $KFDB^{ACC}$ gradually increases before stabilizing. Although there are some fluctuations, the variations are no more than 0.5% overall. Based on this, we conclude that the proposed KFDB effectively reduces the risk of overfitting while improving the model’s expressiveness.
- 3) *The Superiority Over the Classical KDB*: From the results of the last two rows in Tables I and II, we clearly find that, compared to KDB with K values ranging from 1 to 5, $KFDB^{MSE}$ can achieve better ACC on 41, 41, 49, 50, and 50 datasets, significantly outperforms on 22, 28, 30, 30, and 33 datasets, and only loses on 2, 1, 1, 1, and 1 datasets, respectively. At the same time, $KFDB^{ACC}$ also achieves similar results, which obtains better ACC on

TABLE V
SUMMARY TEST RESULTS FOR KFDB^{MSE} VERSUS ITS COMPETITORS

	KFDB ^{MSE}	DL-FSKDB	KDF	KDB	STAN	TAN	WANBIA	NB	RF	LR
KFDB ^{MSE}	-	18 (2)	21 (3)	13 (3)	17 (2)	10 (2)	25 (6)	20 (7)	18 (7)	26 (8)
DL-FSKDB	41 (18)	-	36 (8)	30 (2)	33 (10)	25 (4)	26 (10)	20 (9)	31 (11)	28 (15)
KDF	37 (24)	23 (1)	-	16 (5)	30 (8)	15 (3)	28 (10)	22 (11)	29 (6)	27 (17)
KDB	43 (24)	29 (3)	41 (15)	-	34 (10)	21 (5)	27 (11)	23 (12)	33 (9)	28 (21)
STAN	41 (13)	26 (8)	28 (10)	25 (7)	-	24 (6)	28 (13)	28 (16)	27 (9)	27 (14)
TAN	45 (25)	34 (2)	43 (16)	35 (10)	34 (12)	-	31 (14)	24 (15)	33 (11)	30 (18)
WANBIA	35 (23)	34 (20)	32 (21)	33 (20)	32 (22)	29 (19)	-	25 (4)	31 (21)	24 (9)
NB	38 (29)	40 (24)	38 (28)	35 (27)	32 (26)	34 (24)	35 (15)	-	29 (24)	29 (19)
RF	41 (19)	29 (13)	30 (14)	27 (13)	32 (11)	26 (13)	29 (17)	31 (12)	-	31 (14)
LR	34 (27)	32 (23)	33 (24)	32 (24)	33 (24)	30 (22)	36 (20)	31 (22)	29 (23)	-

TABLE VI
SUMMARY TEST RESULTS FOR KFDB^{ACC} VERSUS ITS COMPETITORS

	KFDB ^{ACC}	DL-FSKDB	KDF	KDB	STAN	TAN	WANBIA	NB	RF	LR
KFDB ^{ACC}	-	17 (2)	22 (8)	18 (5)	22 (4)	15 (5)	23 (6)	20 (4)	21 (6)	26 (15)
DL-FSKDB	42 (14)	-	36 (8)	30 (2)	33 (10)	25 (4)	26 (10)	20 (9)	31 (11)	28 (15)
KDF	37 (16)	23 (1)	-	16 (5)	30 (8)	15 (3)	28 (10)	22 (11)	29 (6)	27 (17)
KDB	38 (19)	29 (3)	41 (15)	-	34 (10)	21 (5)	27 (11)	23 (12)	33 (9)	28 (21)
STAN	37 (10)	26 (8)	28 (10)	25 (7)	-	24 (6)	28 (13)	28 (16)	27 (9)	27 (14)
TAN	41 (21)	34 (2)	43 (16)	35 (10)	34 (12)	-	31 (14)	24 (15)	33 (11)	30 (18)
WANBIA	36 (22)	34 (20)	32 (21)	33 (20)	32 (22)	29 (19)	-	25 (4)	31 (21)	24 (9)
NB	37 (26)	40 (24)	38 (28)	35 (27)	32 (26)	34 (24)	35 (15)	-	29 (24)	29 (19)
RF	39 (13)	29 (13)	30 (14)	27 (13)	32 (11)	26 (13)	29 (17)	31 (12)	-	31 (14)
LR	34 (29)	32 (23)	33 (24)	32 (24)	33 (24)	30 (22)	36 (20)	31 (22)	29 (23)	-

TABLE VII
RANKING TEST RESULTS FOR KFDB^{MSE} VERSUS ITS COMPETITORS

	Wins—Losses	Wins	Losses
KFDB ^{MSE}	162	202	40
KDF	54	139	85
STAN	29	125	96
DL-FSKDB	9	96	87
KDB	1	111	110
RF	-5	121	126
TAN	-25	98	123
WANBIA	-43	116	159
LR	-74	135	209
NB	-108	108	216

TABLE VIII
RANKING TEST RESULTS FOR KFDB^{ACC} VERSUS ITS COMPETITORS

	Wins—Losses	Wins	Losses
KFDB ^{ACC}	115	170	55
KDF	67	144	77
STAN	34	127	93
DL-FSKDB	13	96	83
KDB	8	113	105
RF	0	120	120
TAN	-18	101	119
WANBIA	-42	116	158
LR	-69	142	211
NB	-108	105	213

30, 36, 44, 45, and 48 datasets, significantly wins on 12, 21, 25, 33, and 36 datasets, and loses on 4, 3, 5, 5, and 3 datasets, respectively. These results conclusively demonstrate that KFDB performs significantly better than KDB.

- 4) *The Significance on the Wilcoxon Signed-Rank Test:* According to the results of the Wilcoxon signed-rank test presented in Tables III and IV, regardless of the significance level (either $\alpha = 0.05$ or $\alpha = 0.1$), our proposed KFDB^{MSE} and KFDB^{ACC} significantly outperform the corresponding KDB in terms of classification ACC.

After comparing the results with the classical KDB model, we further evaluated KFDB against several closely related state-of-the-art models, including DL-FSKDB, KDF, KDB, STAN, TAN, WANBIA, NB, RF, and LR. To ensure fairness, the value of K in these models was set to the value

that achieved the best performance in previous experiments: KFDB^{MSE} (K = 3), KFDB^{ACC} (K = 4), and KDB (K = 1). Due to page limitations, here we only present the summary test results for KFDB^{MSE} and KFDB^{ACC} in Tables V and VI. In these tables, for each entry $i(j)$, i indicates the number of datasets where the model in the corresponding column outperforms the model in the corresponding row in terms of classification ACC, while j indicates the number of datasets where the model in the column achieves a significant improvement over the model in the row, with a significance level of $p = 0.05$.

Tables V and VI, we observe that KFDB^{MSE} outperforms DL-FSKDB (18 wins and 2 losses), KDF (24 wins and 3 losses), KDB (24 wins and 3 losses), STAN (13 wins and 2 losses), TAN (25 wins and 2 losses), WANBIA (23 wins and 6 losses), NB (29 wins and 7 losses), RF (19 wins and 7 losses), and LR (27 wins and 8 losses). Similarly, KFDB^{ACC} surpasses DL-FSKDB (14 wins and 2 losses), KDF (16 wins and 8 losses), KDB (19 wins and 5 losses), STAN (10 wins and 4 losses), TAN (21 wins and 5 losses), WANBIA (22 wins and 6 losses), NB (26 wins and 4 losses), RF (13 wins and 6 losses), and LR (29 wins and 15 losses). Furthermore, Tables VII and VIII present the ranking test results for KFDB^{MSE} and KFDB^{ACC}, respectively, where it is evident that both KFDB^{MSE} and KFDB^{ACC} consistently rank first. Finally, the Wilcoxon signed-rank test results in Tables IX and X also confirm that KFDB^{MSE} and KFDB^{ACC} significantly outperform all the other models in the comparison, regardless of whether the significance level is $\alpha = 0.05$ or $\alpha = 0.1$. In summary, our proposed KFDB demonstrates a clear and consistent superiority over all the existing state-of-the-art models used for comparison.

C. Structural Analysis

To thoroughly analyze the structural flexibility of KFDB, we further propose another evaluation metric called dependency arc ratio to measure the proportion of the total number of

TABLE IX
CLASSIFICATION ACC COMPARISONS OF THE WILCOXON TEST FOR KFDB^{MSE} VERSUS ITS COMPETITORS

	KFDB ^{MSE}	DL-FSKDB	KDF	KDB	STAN	TAN	WANBIA	NB	RF	LR
KFDB ^{MSE}	-	o	o	o	o	o	o	o	o	o
DL-FSKDB	•	-	•	•				•		•
KDF	•		-	•		•		•		•
KDB	•		•	-				•		•
STAN	•		•		-					
TAN	•		•			-				
WANBIA	•		•				-			•
NB	•	•	•					-		
RF	•		•						-	
LR	•									-

TABLE X
CLASSIFICATION ACC COMPARISONS OF THE WILCOXON TEST FOR KFDB^{ACC} VERSUS ITS COMPETITORS

	KFDB ^{ACC}	DL-FSKDB	KDF	KDB	STAN	TAN	WANBIA	NB	RF	LR
KFDB ^{ACC}	-	o	o	o	o	o	o	o	o	o
DL-FSKDB	•	-	•	•				•		•
KDF	•		-	•		•		•		•
KDB	•		•	-				•		•
STAN	•		•		-					
TAN	•		•			-				
WANBIA	•		•				-			•
NB	•	•	•					-		
RF	•		•						-	
LR	•									-

dependency arcs in KFDB relative to those in KDB, which can be represented by the following equation:

$$l = \frac{\sum_{j=1}^m e_j}{\sum_{j=1}^m \min(j-1, K)} \quad (18)$$

where l is the dependency arc ratio, m is the number of attributes, e_j is the number of dependency arcs of the j th attribute in KFDB, and K is the maximum number of dependency arcs in KDB.

We present the scatter plot in Fig. 2 to illustrate the relationship between the number of attributes, the dependency arc ratios, and classification performance. Specifically, in Fig. 2, the X - and Y -axes represent the number of attributes and the dependency arc ratio, respectively. The different colored shapes represent the classification ACC comparison results for KFDB and KDB. Additionally, the average dependency arc ratio across 60 datasets is represented by a red dashed line. As indicated by (18), a lower value of l corresponds to a higher proportion of eliminated dependency arcs.

From Fig. 2, we draw two main conclusions.

- 1) As the number of attributes increases, the ratio of eliminated dependency arcs generally rises. For example, when we divide the 60 datasets into low-dimensional datasets (fewer than 40 attributes) and high-dimensional datasets (40 or more attributes), we observe that in Fig. 2(c), 28 out of the 48 low-dimensional datasets have a dependency arc ratio of less than 0.4, accounting for 58%. Among the remaining 12 high-dimensional datasets, 9 of them have a dependency arc ratio of less than 0.4, representing 75%. In KFDB^{MSE} ($K = 5$), similar trends are observed: 69% of the low-dimensional datasets have a dependency arc ratio of less than 0.4, while the proportion for high-dimensional datasets reaches 75%. Moreover, we conclude that KFDB generally outperforms KDB in terms of classification ACC on both low-dimensional and high-dimensional datasets.

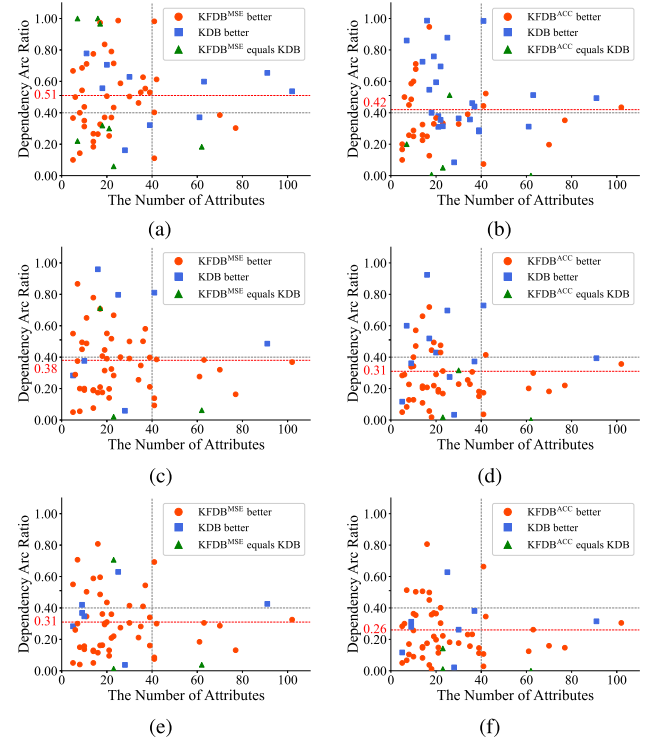


Fig. 2. Relationship between the number of attributes, the dependency arc ratios, and classification performance for KFDB^{MSE} and KFDB^{ACC} at different values of K . (a) KFDB^{MSE} with $K = 1$. (b) KFDB^{ACC} with $K = 1$. (c) KFDB^{MSE} with $K = 3$. (d) KFDB^{ACC} with $K = 3$. (e) KFDB^{MSE} with $K = 5$. (f) KFDB^{ACC} with $K = 5$.

- 2) As the value of K increases, classification performance tends to improve, and the value of l generally decreases. Specifically, for KFDB^{MSE}, the average dependency arc ratio is 0.51 when $K = 1$, and it decreases to 0.26 when $K = 5$. Similarly, for KFDB^{ACC}, the average dependency arc ratio is 0.48 when $K = 1$, and it drops to 0.31 when $K = 5$. In addition, we observe a continued increase in the number of orange circle symbols, while the number

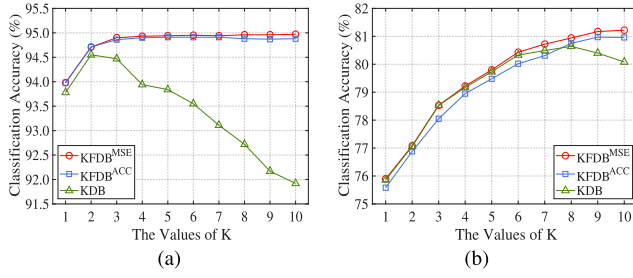


Fig. 3. Classification ACC results for KFDB^{MSE} and KFDB^{ACC} versus KDB on the two large datasets. (a) Census-Income. (b) Connect-40 opening.

TABLE XI
DESCRIPTIONS OF THE TWO LARGE DATASETS WE USED

Dataset	Instance Number	Attribute Number	Class Number
Census-Income	299285	41	2
Connect-40opening	67557	42	3

of blue rectangle and green triangle symbols decreases, further demonstrating that KFDB, compared to KDB, achieves progressively better classification performance as K increases.

D. Large Dataset Analysis

In the current era of big data, the scale of datasets is growing rapidly. Developing algorithms capable of processing large datasets efficiently has become a highly valuable research topic. In this section, we selected two additional datasets from the UCI repository to evaluate the performance of our proposed KFDB in large-scale data applications. The selected large datasets include one binary dataset and one multiclass dataset, each containing over 60 000 instances. The detailed descriptions of them are provided in Table XI. Since large datasets often exhibit more complex conditional dependencies between attributes, it is hypothesized that using a larger K value may improve the model performance. Therefore, we evaluated the classification ACC of KFDB and KDB for K values ranging from 1 to 10. The detailed results are presented in Fig. 3.

From these results, we draw two main conclusions.

- 1) For the binary classification dataset “Census-Income,” KFDB consistently outperforms KDB at the corresponding K values. Meanwhile, we find that on this dataset, KDB achieves the highest classification ACC when $K = 2$, then as K increases, its ACC steadily decreases. Different from that, the classification ACC of both KFDB^{MSE} and KFDB^{ACC} gradually increases before stabilizing, achieving the best result when $K = 10$. Moreover, as K increases, the performance gap between KFDB and KDB widens. This can be attributed to the ability of KFDB to learn the adaptive number of parent nodes for each attribute, which can effectively remove redundant parent attributes. As a result, setting larger values of K usually has minimal impact on classification performance.
- 2) For the multiclass dataset “Connect-40opening,” KDB achieves the best performance when $K = 8$, suggesting that the dataset has highly complex conditional

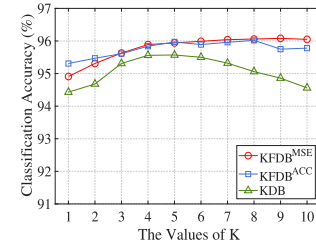


Fig. 4. Classification ACC results for KFDB^{MSE} and KFDB^{ACC} versus KDB on the “noman” dataset.

dependencies between attributes. Although in this case, the ACC of KFDB^{ACC} is comparable to or slightly worse than KDB when $K \leq 8$, when K increases to 9 and 10, both KFDB^{MSE} and KFDB^{ACC} outperform the corresponding KDB. Since determining the optimal K is a challenging task, particularly in the absence of sufficient prior knowledge, these results clearly highlight the advantage of KFDB, which can effectively tackle this challenge by setting a larger K . In summary, both KFDB^{MSE} and KFDB^{ACC} are well-suited for large-scale classification problems, making them highly attractive in the big data era.

E. High-Dimensional Dataset Analysis

Beyond the challenges posed by large-scale datasets, the effective handling of complex high-dimensional data (typically involving more than 100 attributes) is another critical research issue in contemporary data mining. To this end, we selected the “noman” dataset from the UCI repository, which contains 119 attributes, 34 465 instances, and 2 classes to evaluate the performance of KFDB against the classical KDB across values of K ranging from 1 to 10. The detailed results are presented in Fig. 4. As shown, KFDB consistently outperforms KDB in terms of classification ACC. Moreover, the ACC of KFDB exhibits a steady improvement with increasing K values before stabilizing, whereas KDB attains its peak ACC at $K = 4$, followed by a substantial degradation. These findings further demonstrate that KFDB can effectively eliminate redundant attribute dependencies, thereby achieving superior performance on high-dimensional datasets.

Finally, we performed an additional set of experiments for runtime comparisons. We compared the detailed training and classification times (in seconds) of KFDB^{MSE} and KFDB^{ACC} against those of KDB on the “noman” dataset. Based on the results presented in Fig. 5, we derive the following three conclusions.

- 1) As the value of K increases, both training and classification times for all three models exhibit a growing trend. This growth is inevitable, since larger K values necessitate a more exhaustive search for attribute dependencies.
- 2) From Fig. 5(a), we observe that the training times of KFDB^{MSE} and KFDB^{ACC} are only marginally higher than that of KDB. This efficiency can be attributed to the incremental LOOCV strategy employed in KFDB, which requires only one additional pass through the training dataset.

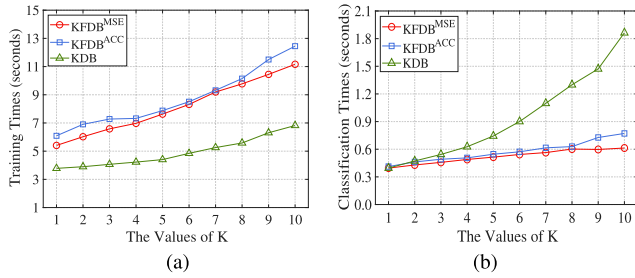


Fig. 5. (a) Training and (b) classification time comparison results for KFDB^{MSE} and KFDB^{ACC} versus KDB on the “noman” dataset.

3) As evidenced in Fig. 5(b), the growth rate of classification time in KDB is higher than that of KFDB. This is mainly because KFDB simplifies the network structure by removing redundant attribute dependencies, making probability estimation simpler and more efficient. In summary, although the training times of KFDB are slightly longer than those of KDB, the classification times are substantially reduced, particularly at larger K values, demonstrating that the proposed KFDB remains computationally efficient.

Moreover, these findings provide us with some guidance on selecting an appropriate K value across different application scenarios. When substantial prior knowledge of the underlying attribute dependencies is available, adopting a smaller K is preferable, as it ensures strong predictive performance while mitigating excessive computational overhead. Conversely, in situations where such prior knowledge is limited, employing a relatively larger K can broaden the search space and thereby yield more competitive classification ACC.

V. CONCLUSION

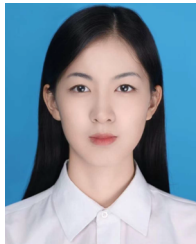
In this study, we identify two limitations of the classical KDB model. Increasing its K value may raise the risk of overfitting, while fixing its K value may constrain the model expressiveness. To address these issues, we propose KFDB classifiers. In KFDB, we first learn multiple candidate submodels based on a pre-set K value; then, we sequentially evaluate them using incremental LOOCV, and finally, we learn an adaptive number of parent nodes for each attribute by minimizing the MSE or maximizing the classification ACC. Extensive experiments on 60 UCI datasets demonstrate the superiority of KFDB. Finally, we perform some in-depth discussions to elucidate some key characteristics of KFDB.

However, KFDB selects the local best submodel for each attribute in a sequential manner, which can be regarded as a somewhat greedy strategy. Expanding the search space is therefore a key direction for future work. Additionally, leveraging graph clustering techniques [39], [40] to infer attribute dependencies and automatically determine the optimal number of parent nodes for each attribute is another interesting direction for our future research.

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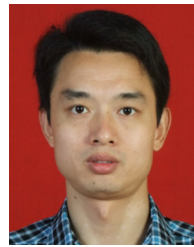
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